



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	7.28 / 10
2022	Class XII (TSBIE)	Narayana Junior College	98.7%
2020	Class X (TSBIE)	Residential School	10 / 10

INTERNSHIPS AND PROJECTS

VLDP INTERN VEDANTA Raipur	May'25 - Jul'25
- Standardized the Anode Change Procedure across 288 reduction pots, achieving 100% checklist-driven execution and improving process - Performed SCADA-based root cause analysis, reducing pot noise and Anode Effects while improving current efficiency and pot stability - Proposed FCN &SHE implementation with 31.7 Cr CAPEX (11 lakh/pot) to enable predictive monitoring and safer, automated operations - Projected 40-140C lower shell temperature, 30-40% lower compressed-air use, 60-80% fewer red-shell events and 1.52.5% higher production - Built a business case projecting 40-60 Cr annual savings, 25.6% ROI, and 18-30 months payback through the improved operational efficiency	
Project Trainee Rashtriya Ispat Nigam Limited (RINL) Vizag	May'24- Jun'24
- Analyzed the theoretical dephosphorization process in Basic Oxygen Furnace (BOF) steelmaking to enhance steel quality and process efficiency - Investigated effects of slag basicity, FeO content, and bath temperature on phosphorus removal, emphasizing targeted process optimization - Developed data-driven strategies to optimize dephosphorization rates and improve steel production quality and high operational effectiveness - Acquired practical hands-on knowledge of integrated steel plant operations, including key units such as BF, Sinter Plant & Steel Melting Shop	
Closed-Loop Recycling of Lithium-Ion Batteries Guided by: Prof. Manish Kumar Kar, IIT Kharagpur	July'25 - Nov'25
- Developed a hydrometallurgical route for NMC black mass using ammonium-sulfate-driven co-crystallization, achieving >98% Ni &Co recovery - Optimized NH:TM ratio, slurry density, temperature and reaction time to improve selective NiMnCo recovery and overall process efficiency - Characterized recovered sulfates using XRD, XRF, SEM, evaluating phase composition, morphology, elemental distribution &thermal stability - Regenerated NMC622 cathode material from recovered sulfates &evaluated electrochemical cycling performance against reference cathodes	
Recycling of Lithium Ion Batteries Guided by: Prof.Chenna Rao Borra, IIT Kharagpur	July'25 - Nov'25
- Executed a 6-stage recycling workflow on 2 spent Li-ion cells, from the initial discharging to material characterization with safe handling - Recovered 14.36 g of black mass and 1.15 g of graphite through a single NaOH-based process for effective binder and collector separation - Demonstrated a scalable, low-energy recycling method enabling efficient recovery of 3 critical metals for circular-economy applications - Confirmed composition via CS analysis and SEM up to 5000, reporting 34.26% carbon and <0.05% sulfur with minimal electrolyte residue	
Life cycle assessment Guided by: Prof.Chenna Rao Borra, IIT Kharagpur	Jan'25 - May'25
- Built an OpenLCA model for 1 kg NMC-622 leaching using the ReCiPe 2016 Midpoint (H) methodology to quantify environmental impacts - Evaluated overall environmental performance, quantifying 2.14 kg CO₂-eq emissions, 0.279 kg 1,4-DCB eq, and 0.032 m water consumption - Identified electricity as the largest environmental hotspot through contribution analysis, enabling data-driven sustainability improvements - Proposed lifecycle optimization strategies to reduce emissions & improve the sustainability of lithium-ion battery recycling using LCA principles	

SKILLS AND EXPERTISE

Technical Core: Optical Microscope | SEM | TEM | XRD analysis | XRF **Testing Softwares:** Xpert High Score | MATLAB | Jupiter | OriginLab

POSITIONS OF RESPONSIBILITY

Senior Photographer ClickKGP IIT Kharagpur	Aug'24 - Present
- Managed content strategy & delivered 500+ assets across 10+ client engagements, driving to 150K+ impressions & growth to 7K+ followers - Led 6+ photography workshops and mentored 50+ members on photography, editing, and content strategy while covering 5+ major events	
General Secretary Sports and Games RP Hall	Jul'24 - Jul'25
- Led key sports initiatives for 850+ boarders, organizing 3 new events and managing an INR 4.5M budget to drive participation and growth - Spearheaded a 42-member sports team across 25+ events; optimized the budget, increasing participation 38% & earning Gold in GC 2024-25	
Head Publicity and Marketing Shaurya	Jul'24 - Jul'25
- Introduced a contingent-based registration & payment system, reducing wait time by 95% & streamlining onboarding across 100+ activities - Led a 20-member team for the Shaurya 2024, delivering 200+ social media posts, engaging 4,500+ students, and supporting 25+ fest events	

EXTRA CURRICULAR ACTIVITIES

Sports: Secured five Gold medals in volleyball at Open IIT and General Championship (25), representing RPH among 21 Halls at IIT Kharagpur Tech: Earned a Silver medal in the Case Study Competition (24) among 21 Halls, organized by Technology Students Gymkhana, IIT Kharagpur Cultural: Orchestrated in Gold-medal winning performances in Illumination and Rangoli for Rajendra Prasad Hall of Residence among 21 Halls Welfare: Mentored 5 junior students under the SWG program, providing comprehensive guidance across academic &non-academic challenges	
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COURSEWORK INFORMATION

Materials Processing | Materials Characterization | XRD and TEM | Computer Application in Metallurgical Processing | Extractive Metallurgy | Phase Transformation and Heat Treatment of Materials | CFT | Corrosion | Mechanical Working of Materials | Iron Making and Steel Making | Probability and Statistics | Transform Calculus | Linear Algebra and Numerical Analysis | Partial Differential Equations | Sustainable Extractive Metallurgy | Stainless Steel and Advanced Ferrous Alloys | Grain Boundary Engineering | DIY Lab | Basic Engineering Mechanics |